EMERSON UNIVERSITY OF MULTAN

Assignment task:

Introduction of Packet tracer and Wireshark

Submitted To:

Dr. Shahzad Bhatti

Subject:

Computer Networking

Submitted From:

Ali Muzamil

Roll No:

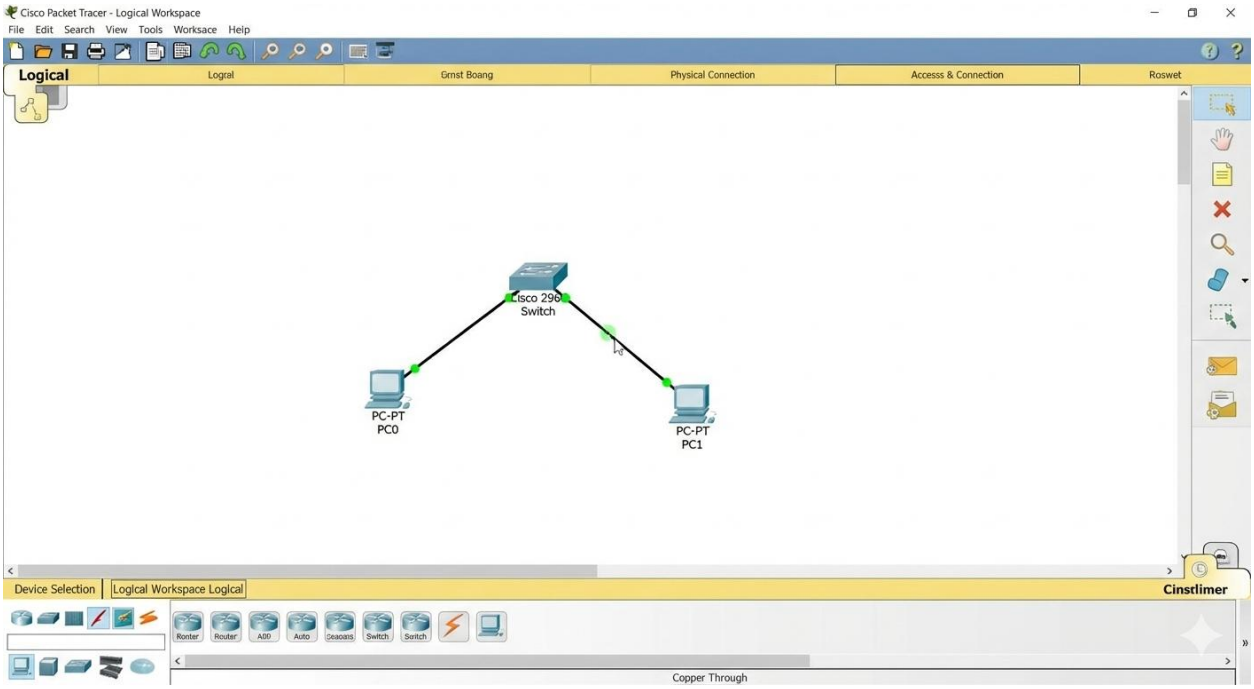
05

BSCS-Evening (B)-4th semester

Section:

2024-2028

Packet Tracer



Introduction

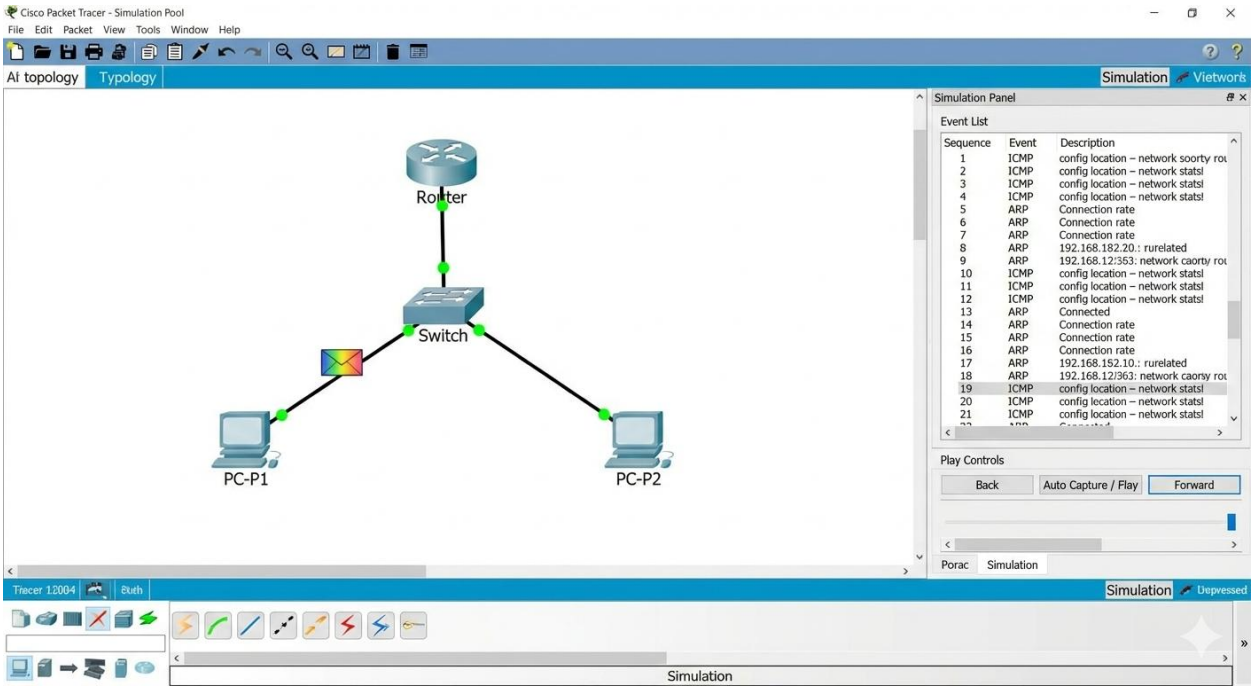
In modern computer networking education, practical exposure is essential for understanding how networks operate, how data flows, and how configurations affect communication. However, building physical labs with routers, switches, and servers is costly and often impractical for students. To address this challenge, **Packet Tracer**, developed by Cisco Systems, provides a powerful simulation environment that enables learners to design, configure, and analyze networks virtually.

Packet Tracer bridges the gap between theoretical networking concepts and real-world implementation by allowing users to visualize packet flow, test configurations, and troubleshoot network issues in a safe, controlled environment. It is widely used in networking courses, especially in Cisco Networking Academy programs, to support hands-on learning without requiring physical hardware.

Historical Development and Purpose

Packet Tracer was first released in **2003** by Cisco Systems to support students enrolled in the Cisco Networking Academy. The primary objective was to create an interactive learning platform where users could simulate real networking scenarios.

Over time, the software evolved significantly. New versions introduced support for Internet of Things (IoT) devices, automation tools, multiuser collaboration, and enhanced protocol simulation. Today, Packet Tracer is not only an educational tool but also a foundational platform for beginners preparing for professional networking certifications such as CCNA.



Concept and Working Environment

Packet Tracer is a **network simulation tool** that models the behavior of networking devices and protocols. Instead of relying on physical equipment, it uses logical models to emulate routers, switches, end devices, and communication protocols.

The working environment consists of:

- A **workspace** where network topology is designed.
- A **device panel** containing routers, switches, PCs, servers, and IoT devices.
- A **configuration interface** supporting both graphical settings and Cisco CLI commands.
- A **simulation engine** that visualizes packet movement.

This environment allows users to construct networks ranging from simple LANs to complex multi-network infrastructures.

Functional Operation of Packet Tracer

To understand how Packet Tracer operates, consider the process of sending data between two computers in a simulated network:

1. The user creates a network topology by placing devices such as PCs, switches, and routers.
2. Each device is configured with IP addresses and necessary protocols.
3. A data packet is generated from the source device.
4. The packet is forwarded by switches using MAC addresses.
5. Routers determine the best path using IP routing tables.
6. The destination device receives and processes the packet.

This process can be observed in **Simulation Mode**, where each step is displayed in sequence, helping learners understand how data travels across a network.

Basic Local Area Network Topology

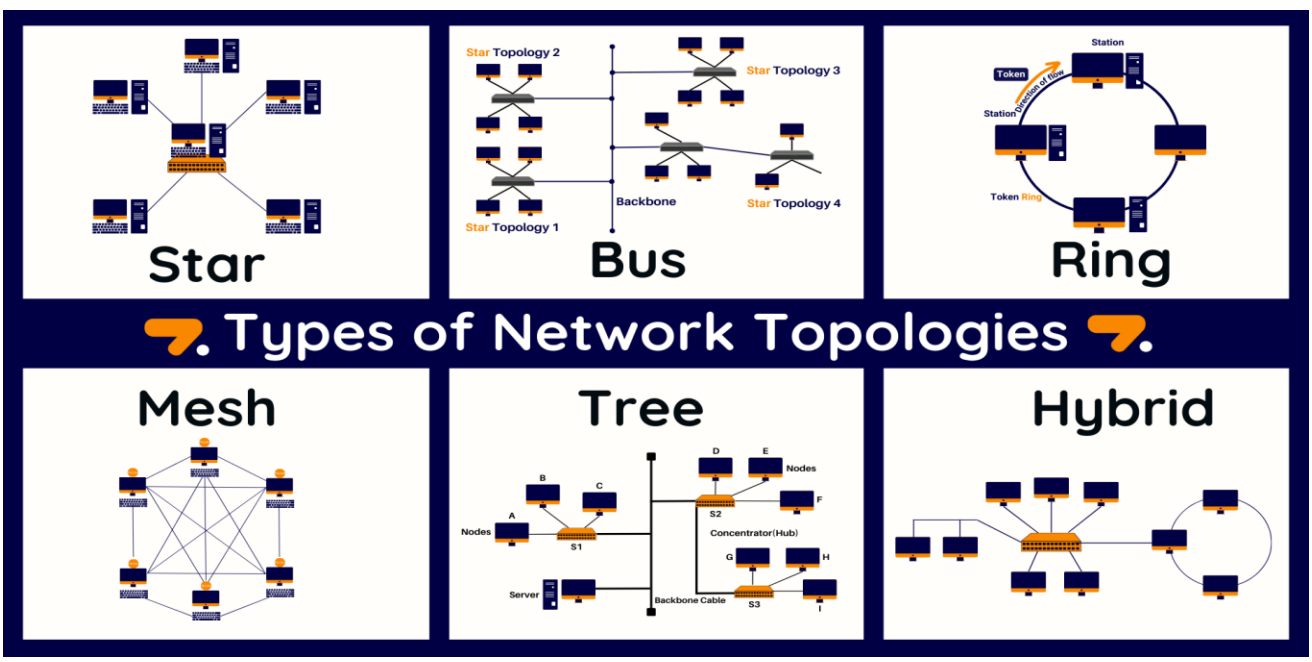


Figure 1: Basic Local Area Network Topology

Explanation:
This figure illustrates a simple LAN where two computers communicate through a switch or some topology types. The router connects the local network to external networks, while the server provides services such as web hosting or file sharing.

Packet Flow in Simulation Mode

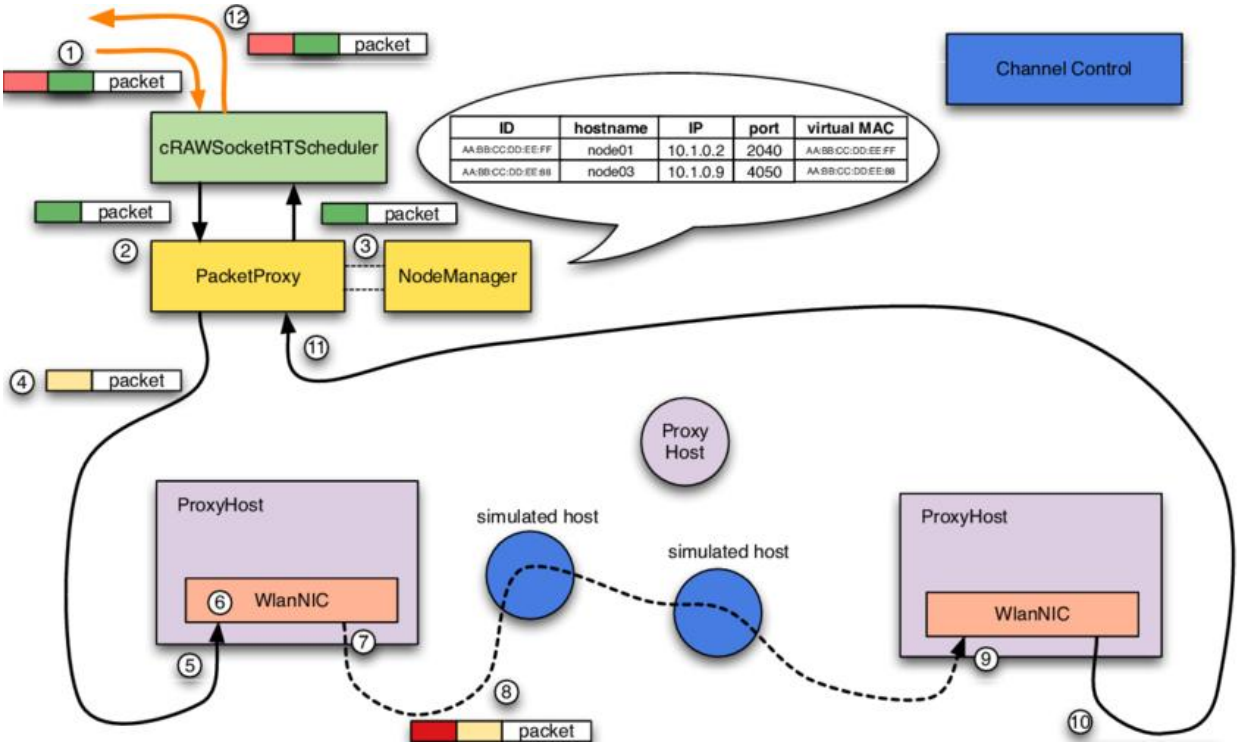


Figure 2: Packet Flow in Simulation Mode

Explanation:
This diagram shows how packets move inside a network simulation system.
A scheduler controls when packets are sent, while a packet proxy forwards them between simulated and real network parts.
The node manager keeps a table of device identities, IP addresses, ports, and virtual MAC addresses to ensure correct routing.
Proxy hosts with wireless network interfaces act as bridges connecting simulated hosts to the network.The dotted lines represent virtual links, illustrating how simulated devices communicate under central channel control.

Educational Significance

Packet Tracer plays a vital role in networking education because it transforms abstract concepts into observable processes. Students can see how routing decisions are made, how switching works, and how protocols interact.

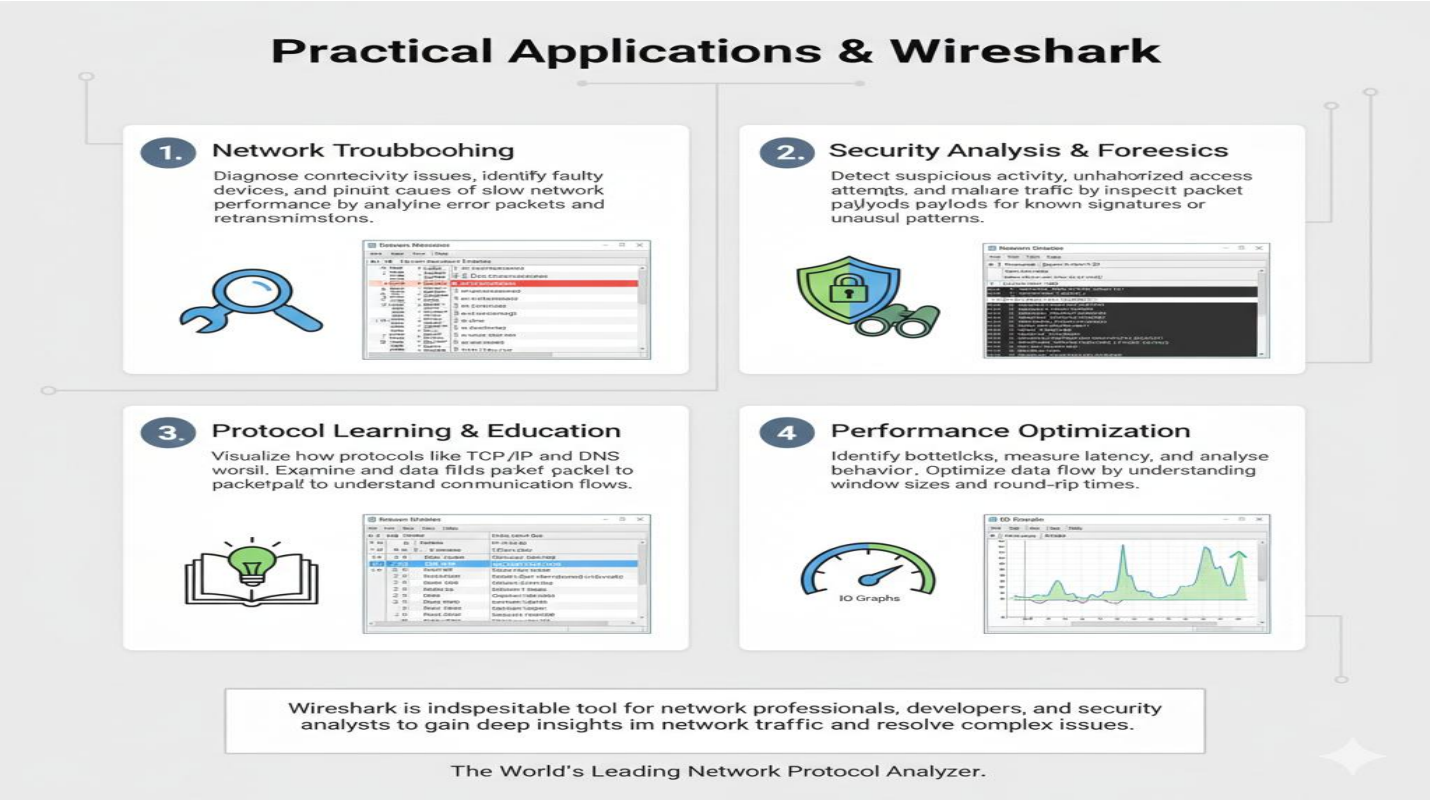
The simulation environment encourages experimentation. Learners can intentionally misconfigure devices to observe failures and then apply troubleshooting techniques. This trial-and-error approach strengthens problem-solving skills and deepens conceptual understanding.

Furthermore, the inclusion of Cisco's command-line interface allows students to practice real device configuration commands, preparing them for professional environments.

Practical Applications

Packet Tracer is used in various academic and training contexts. In university labs, it helps students design and test network architectures. In certification preparation, it enables candidates to practice routing, switching, VLAN configuration, and troubleshooting. The tool is also used to simulate IoT systems such as smart homes, where sensors, controllers, and network devices interact.

By enabling safe testing and rapid prototyping, Packet Tracer allows users to evaluate network designs before implementing them in real-world environments.



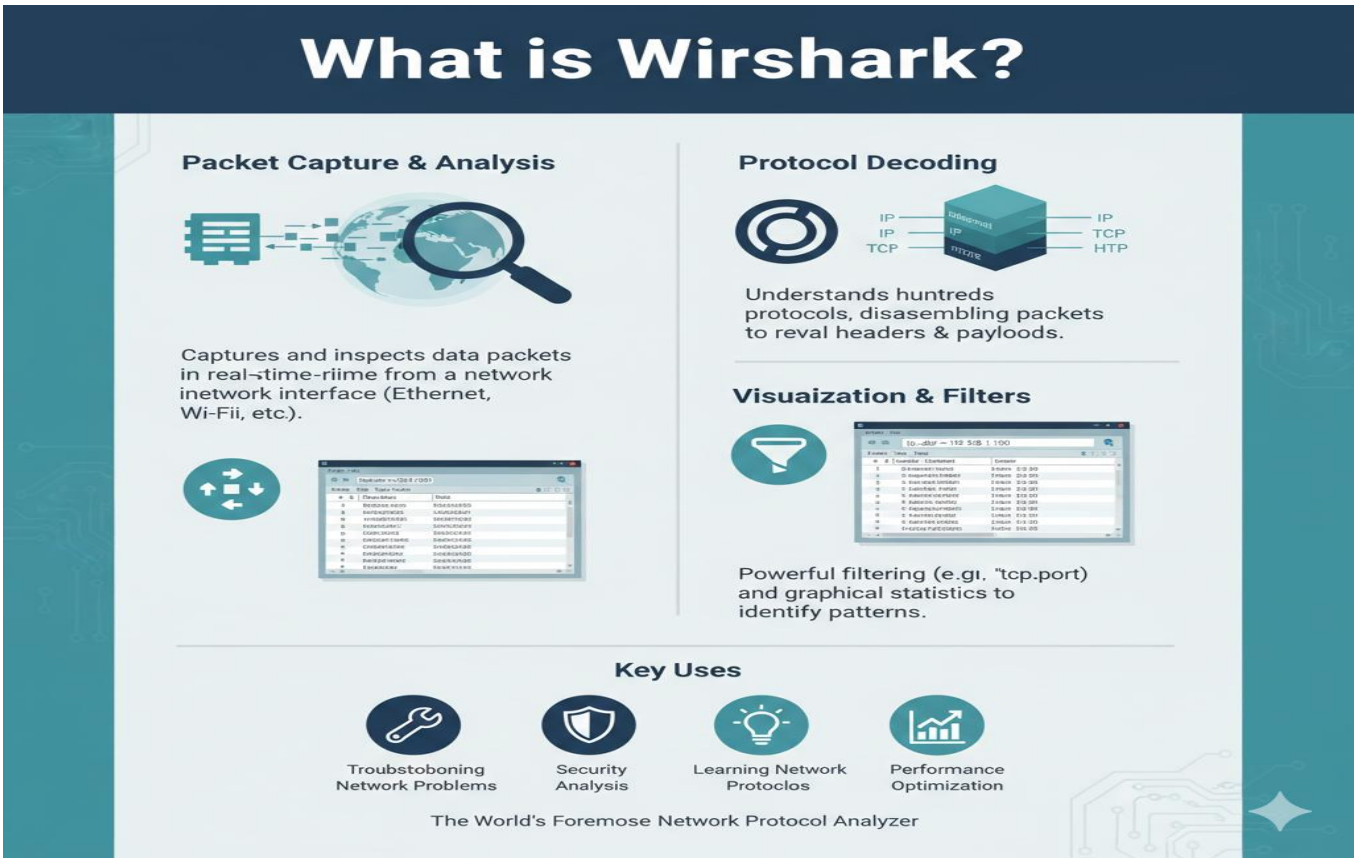
Limitations of Packet Tracer

Despite its strengths, Packet Tracer has certain limitations. It does not fully replicate the performance characteristics of real hardware, and some advanced enterprise features are not supported. Additionally, while it accurately models many protocols, it simplifies certain behaviors for educational purposes.

Therefore, while Packet Tracer is excellent for learning and initial design, real-world deployment requires experience with physical devices and production-grade environments.

Through visualization and hands-on experimentation, Packet Tracer enhances conceptual understanding and prepares learners for real-world networking challenges. Although it cannot fully replace real hardware experience, it remains an indispensable tool for building foundational skills in computer networking.

Wireshark



Introduction

Computer networks form the backbone of modern communication systems, enabling data exchange across local and global infrastructures. Understanding how data moves through a network requires tools that provide visibility into packet transmission, protocol behavior, and device interactions. **Wireshark** is a widely used network protocol analyzer that captures and examines real-time network traffic, allowing users to observe how communication occurs at a granular level.

Originally developed in 1998 by Gerald Combs, Wireshark has evolved into an industry-standard tool for network troubleshooting, cybersecurity analysis, and academic research. In contrast, **Packet Tracer**, developed by Cisco Systems, provides a simulated environment for designing and testing networks. Together, these tools support both conceptual learning and real-world network analysis.

What Wireshark Is and Why It Matters

Wireshark is a packet capture and analysis tool that allows users to see the actual data packets traveling across a network. Instead of relying on assumptions about network behavior, it shows real communication between devices.

When a device sends data—such as opening a website or sending an email—the information is divided into packets. Wireshark captures these packets and displays details such as:

- Source and destination addresses
- Protocols used (HTTP, DNS)
- Packet size and timing
- Errors or retransmissions

This visibility makes Wireshark essential for diagnosing network problems, detecting suspicious activity, and understanding protocol operations.

How Wireshark Works?

In a real network, multiple devices communicate simultaneously. Wireshark monitors this communication by capturing packets through the network interface card (NIC).

Practical Workflow

1. The NIC collects packets from the network.

2. Wireshark records and timestamps each packet.
3. The software decodes protocol layers (Ethernet → IP → TCP → Application).
4. Users filter and analyze traffic to find relevant information.

Real Packet Capture Process

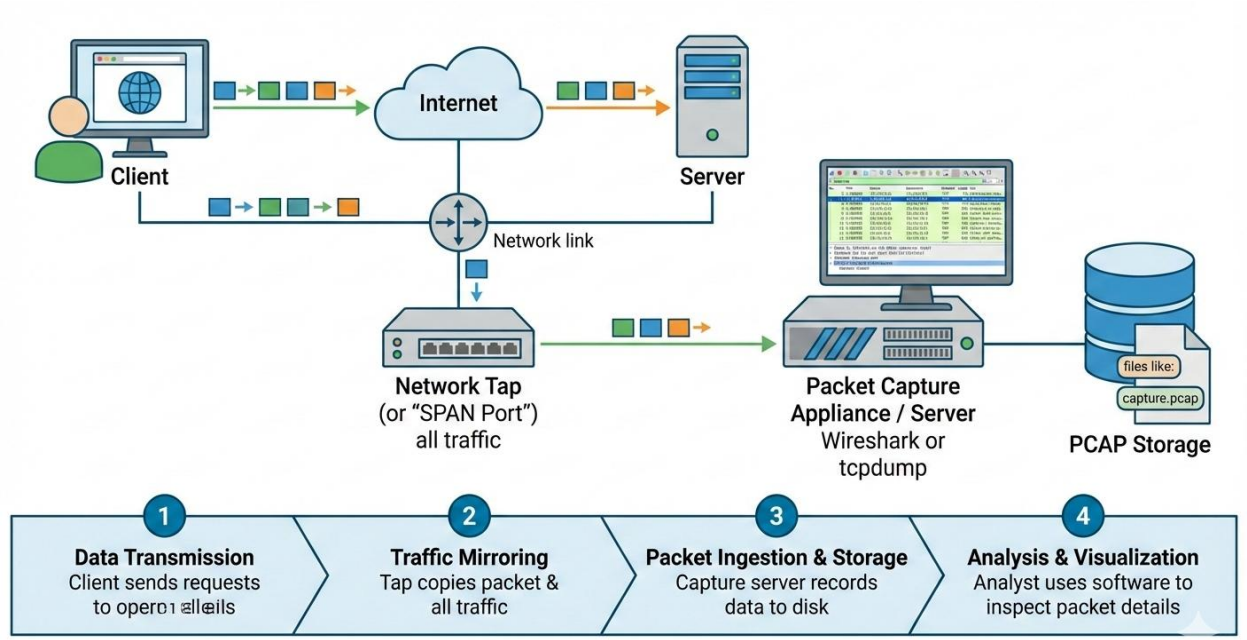


Figure 1: Real Packet Capture Process

User Activity → Network Traffic → NIC Capture → Wireshark Analysis → Problem Detection

This process helps administrators identify issues such as slow connections, failed requests, or unauthorized access.

Understanding Packet Analysis Through an Example

Consider a student opening a website:

1. The computer sends a DNS request to resolve the website name.
2. The DNS server responds with an IP address.
3. The browser sends an HTTP request to the server.
4. The server responds with webpage data.

Wireshark captures each step, allowing the user to see:

- DNS query and response
- TCP handshake (SYN, SYN-ACK, ACK)
- HTTP request and response codes

This step-by-step visibility helps learners understand how everyday internet activities function at the network level.

Educational and Professional Importance

Wireshark is valuable because it connects theory with real network behavior. Students studying protocols often struggle to visualize how communication occurs. By observing real packets, they gain clarity on:

- How TCP establishes reliable connections
- How routing affects packet delivery
- How delays and packet loss impact performance

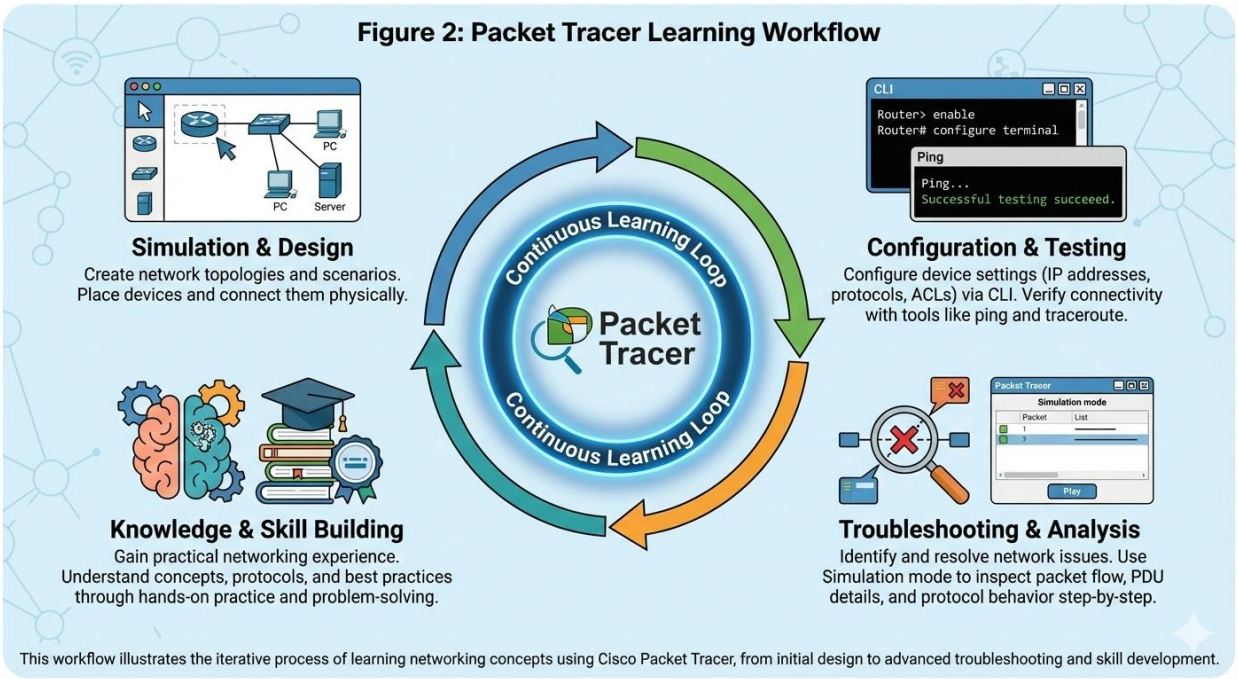
In professional environments, Wireshark is used to:

- Diagnose slow network performance
- Detect malware communication
- Investigate security incidents
- Verify protocol implementations

Packet Tracer: A Simulation-Based Learning Tool

While Wireshark analyzes real traffic, Packet Tracer provides a virtual lab where students can design and configure networks without physical hardware. It simulates routers, switches, and hosts, allowing learners to observe packet flow in a controlled environment.

Packet Tracer Learning Workflow



Network Design → Device Configuration → Simulated Packet Flow → Concept Understanding
Packet Tracer simplifies complex networking concepts by visualizing packet movement and protocol behavior.

Comparison: Wireshark vs Packet Tracer

Wireshark operates in real environments, capturing authentic network traffic and exposing the true complexity of protocol interactions. It is primarily used for analysis, troubleshooting, and security monitoring. Packet Tracer, on the other hand, creates a simulated environment where users can design networks, configure devices, and observe packet flow without real hardware.

From a learning perspective, Packet Tracer is ideal for beginners because it simplifies networking concepts and provides visual feedback. Wireshark becomes more valuable as students advance, enabling them to analyze real-world traffic and understand how theoretical concepts apply in practice. Another key difference is the level of abstraction. Packet Tracer abstracts low-level details to focus on configuration and topology, whereas Wireshark exposes raw packet data, headers, and protocol fields. This makes Wireshark more complex but also more powerful for diagnostics and research.